

Commutative Ring with unity.

Basic Preliminaries: Let R be a Commutative Ring with unity $1 \neq 0$, $A \subseteq R$ is subset, $I \subseteq R$ Ideal.

1). $(A) = RAR$

2). $I = R$ iff I has a unit. (not necessary Commutative)

3) R is field iff only Ideal of R is 0 or R .

4) $P \subseteq R$ is prime $\Leftrightarrow R/P$ is integral domain. $\hookrightarrow \curvearrowright$

5). $M \subseteq R$ is Maximal $\Leftrightarrow R/M$ is field.

Def. Let R be an Integral domain, A map

$$N: R \rightarrow \mathbb{N} \cup \{0\}$$

is called a Norm if $N(0) = 0$. If $N(a) > 0$ for all $a \neq 0$, then it is called positive Norm.

Def. Let R be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1). a is called a multiple of b if there exists $x \in R$ s.t. $a = bx$. Simply denote $b|a$.

2). A non-zero $d \in R$ is called a greatest common divisor of a and b if

$$\begin{cases} d|a \text{ and } d|b \\ d'|a \text{ and } d'|b, \text{ then } d'|d. \end{cases}$$

Since $b|a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$, d is a greatest Common divisor of a and b if

$$(a, b) \subseteq (d) \text{ and } (a, b) \subseteq (d') \Rightarrow (d) \subseteq (d'). \quad \text{Directly,}$$

Prop. $(a, b) = (d) \Rightarrow d$ is g.c.d of a and b .

But the converse is false, In $\mathbb{Z}[x]$, 1 is g.c.d of 2 and x but $R = (1) \neq (2, x)$.

Def. The integral domain R is said to be a Euclidean domain if: there exists a Norm N s.t. for any $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ s.t. $a = qb + r$ where $N(r) < N(b)$ or $r = 0$.

Lemma. If R is an Euclidean domain with a Norm N , then for any $a, b \in R$ with $b \neq 0$, there exists a sequence $\{r_k\}_{k=0}^{\infty}$ satisfying: put $r_{-2} = a$, $r_{-1} = b$,

$$\begin{cases} \text{for } 0 \leq k \leq n+1, r_{k-2} = q_k \cdot r_{k-1} + r_k \text{ for some } q_k \in R, \\ N(b) > N(r_0) > N(r_1) > \dots > N(r_n), \\ r_{n+1} = r_{n+2} = \dots = 0. \end{cases}$$

This $\{r_k\}_{k=0}^{\infty}$ is called remainder sequence for a and b .

Prop. Let R be an Euclidean domain. Then, for any non-zero ideal $I \subseteq R$, $I = (d)$, where d is any non-zero element in I of minimum norm.

Proof. Let $I \subseteq R$ be a non-zero Ideal. observe that: the set

$$N[I \setminus \{0\}] = \{N(a) \mid a \in I, a \neq 0\}$$

has a minimum element $m \in N \setminus \{0\}$ by the Well-ordering principle,

take $d \in N^{-1}[\{m\}]$. Trivially, $(d) \subseteq I$. To show the reverse inclusion, let $a \in I$ be given. Since R possess a Division Algorithm, for some $q, r \in R$,

$$a = qd + r \quad N(r) < N(d) \text{ or } r = 0$$

But, by the Minimality of $N(d)$, r must be zero. Therefore, $a = qd \in (d)$. $\square$

Lemma. Let R be an Integral domain, let $d, d' \in R$.

If $(d) = (d')$, then $d = ud'$ for some unit $u \in R$.

Proof. If either d or d' is zero, there is nothing to prove.

Assume both d and d' are non-zero. Since $d \in (d')$ and $d' \in (d)$,

$$d = xd' \text{ and } d' = yd \text{ for some } x, y \in R.$$

$$\text{Now, } d = xd' = xyd \text{ implies } d(1 - xy) = 0.$$

Since R has no zero divisor and $d \neq 0$, $xy = 1$. x and y both are units. $\square$

Thm. Let R be an Euclidean domain, and a and b be non-zero elements in R .

Let $\{r_i\}$ be a remainder sequence for a and b . Then,

$$(r_n) = (a, b).$$

Proof. Clearly, $r_n | r_{n-1}$ since $r_{n-1} = q_{n-1}r_n$ and $r_n | r_n$.

If r_n divides r_{k+1} and r_k , then r_n divides $r_{k-1} = q_{k+1}r_k + r_{k+1}$,

thus r_n divides r_k and r_{k-1} . Inductively, r_n divides $b = r_{-1}$ and $a = r_{-2}$.

This proves that $(a, b) \subseteq (r_n)$. To show the reverse inclusion,

$a = q_0b + r_0$ implies $r_0 \in (a, b)$ and $b = q_1r_0 + r_1$ implies $r_1 \in (b, r_0) \subseteq (a, b)$.

And, if $r_{k-1}, r_k \in (a, b)$, then $r_{k-1} = q_{k+1}r_k + r_{k+1}$ implies $r_{k+1} \in (a, b)$.

That is, $r_k, r_{k+1} \in (a, b)$. Inductively, $r_n \in (a, b)$. $\square$

Def. An Integral Domain R is called Principal Ideal Domain if:
Every Ideal in R is principal.

Directly, Euclidean domain $\Rightarrow$ P.I.D.

Prop. Let R be a P.I.D., and let a and b be non-zero elements of R .

Let $d \in R$ s.t. $(d) = (a, b)$. Then,

- 1) d is a g.c.d. of a and b
- 2) $d = ax + by$ for some $x, y \in R$
- 3) d is unique up to multiplication by a unit of R .

Prop. Every nonzero prime ideal in a P.I.D is a Maximal.

Proof. Let $(p) \subsetneq R$ be a Prime Ideal, $p \neq 0$. And let $m \in R$ s.t.

$$(p) \subseteq (m) \subseteq R.$$

Enough to show that $(m) = (p)$ or $(m) = R$. By the assumption, $p = rm$ for some $r \in R$.

And, since (p) is prime, $rm \in (p)$ implies $r \in (p)$ or $m \in (p)$.

If $r \in (p)$, then $r = ps$ for some $s \in R$, $p = rm = psm$. Thus $sm = 1$, m is unit, $(m) = R$.

If $m \in (p)$, then $(m) \subseteq (p)$, thus $(m) = (p)$. ~~pro~~

The Converse is true in general integral domain.

Def. Let R be an Integral domain, let $r \in R$ be a nonzero, not a unit.

1). A nonzero, not unit $r \in R$ is called irreducible in R if:

if $r = ab$ with $a, b \in R$, then either a or b is a unit in R .

2). A nonzero $p \in R$ is called prime in R if: $(p) \subseteq R$ is prime ideal.

Equivalently, $p \in R$ is prime $\Leftrightarrow p$ is nonzero, not unit, $p|ab \Rightarrow p|a$ or $p|b$.

Prop. Prime element is irreducible in an integral domain.

Proof. Let $p \in R$ be a prime element in an integral domain.

If $p = ab$, then $p = ab|ab$, thus $p|a$ or $p|b$. Either $p|a$ or $p|b$,

$$\begin{cases} p = prb & \text{if } p|a. \\ p = asp & \text{if } p|b \end{cases}$$

for some $r, s \in R$, which implies $rb = 1$ or $as = 1$, b or a is a unit. $\square$

Prop. Irreducible element in P.I.D is a prime.

Proof. Let $r \in R$ be an irreducible element in a P.I.D.

Suppose that an Ideal $M \subseteq R$ satisfies $(r) \subseteq M \subseteq R$. Since R is P.I.D, let $M = (m)$.

Since $r \in (m)$, $r = mx$ for some $x \in R$. And, r is irreducible, thus either m or x is a unit.

If m is a unit, then $(m) = R$, if x is a unit, then $m = rx^{-1} \in (r)$. Thus $(r) = (m)$.

Therefore, (r) is Maximal, thus prime.

In sum, in P.I.D., $\overset{\text{P.I.D.}}{\text{Maximal}} \Leftrightarrow \overset{\text{P.I.D.}}{\text{Prime}} \Leftrightarrow \text{Irreducible}$.

Def. A Unique Factorization Domain is an integral domain R in which every nonzero, non-unit $r \in R$ satisfies the following two properties:

1. r can be written as a finite product of irreducibles $p_i \in R$. (Not necessary distinct)
2. This representation is unique up to associates.

Prop. In a U.F.D., prime $\Leftrightarrow$ irreducible.

Proof. Generally, prime element is irreducible. Let $p \in R$ be an irreducible in a U.F.D. R .

Assume that $p \mid ab$ for some $a, b \in R$. That is, $ab = pc$ for some $c \in R$.

Since R is U.F.D. and a, b are nonzero, not unit, a and b have factorizations:
(If either a or b is unit, then p also must divide other one)

$$a = u \cdot p_1 \cdots p_n, \quad b = v \cdot q_1 \cdots q_m \quad (u, v \text{ are units, } p_i, q_j \text{ are irreducible})$$

Now, $ab = uv p_1 \cdots p_n q_1 \cdots q_m = pc$

and this factorization is unique up to associate, p must associate ^{to} one of $p_1, \dots, p_n, q_1, \dots, q_m$.

Thus, $p \mid a$ or $p \mid b$.

In sum, in U.F.D., $\text{Maximal} \xrightarrow{?} \text{Prime} \xrightarrow{\text{U.F.D.}} \text{Irreducible}$.

Thm. Every Principal Ideal Domain is Unique Factorization Domain.

Proof. Let R be a P.I.D., and $r \in R$ be a nonzero, not unit.

If r is irreducible, then clear. Suppose r is reducible. Then,

$r = r_1 r_2$ for some nonzero r_1, r_2 . If both r_1 and r_2 are irreducible, clear.

If either r_1 or r_2 is not irreducible, wlog, r_1 is reducible,

$r_1 = r_{11} r_{12}$ for some nonzero r_{11}, r_{12} . Clearly,

$$(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq R$$

because $r \in (r_1)$ since $r = r_1 r_2$, but $r_1 \notin (r)$, since r_2 is not unit.

(specifically, if $r_1 \in (r)$, then $r_1 = r x = r_1 r_2 x \Rightarrow r_1 (1 - r_2 x) = 0 \Rightarrow r_2 x = 1 \Rightarrow r_2$ unit)

Inductively, we can construct an ascending chain of ideals:

$$(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq \dots \subsetneq R$$

But, Every Ascending Chain in P.I.D will be stationary. Because:

Let $\{I_n\}_{n=1}^{\infty}$ be a collection of Ideals of P.I.D R s.t

$$I_1 \subseteq I_2 \subseteq \dots \subseteq R.$$

Denote $I = \bigcup_{n=1}^{\infty} I_n$. Then, it is Ideal, because for any $r \in R$ and $a \in I$, $a \in I_n$ for some $n \in \mathbb{N}$

thus $ra \in I_n \subseteq I$. Also, I is a subring, because $\{I_n\}$ is nested. Thus I is ideal.

Since R is P.I.D, denote $I = (d)$ for some $d \in R$. Since $d \in I_1$, $d \in I_n$ for some $n \in \mathbb{N}$.

This means $I_n \subseteq I = (d) \subseteq I_n$, thus $I_n = I$. $\perp$ P.I.D $\Rightarrow$ Noether Domain.

Therefore there is No infinite strictly ascending chain,

thus r has a finite irreducible factorization. Now the Uniqueness remains.

Using induction for number of irreducible factors for some factorization of r .

If $n=1$, then r is irreducible. If $r = p_1 p_2 \dots p_m$ is other factorization where p_i are irreducible, then r must divide one of p_i , because r is prime. That is, $p_i = u \cdot r$ for some i .

Wlog, put $i=1$. Then, $r = u r p_2 \dots p_m \Rightarrow r(1 - u p_2 \dots p_m) = 0$, but since $p_2, \dots, p_m$ are not units,

it can not be possible. Suppose for $n-1$, the Uniqueness holds. If r has two factorization

$$r = p_1 p_2 \dots p_n = q_1 q_2 \dots q_m \quad m \geq n$$

Similarly, p_1 must divide one of q_i , wlog q_1 . Then $p_2 \dots p_n = u q_2 \dots q_m$, for some unit u .

These are same representation, by the induction.

Def. An integral domain R in which every ideal generated by two elements is principal is called a Bézout domain.

Prop. Let R be an integral domain.

R is Bézout domain if and only if for any $a, b \in R$, $d = \gcd(a, b)$ exists with $d = ax + by$ for some $x, y \in R$.

Proof. Suppose R is Bézout domain. Let $a, b \in R$ be given. Then,

$$(a, b) = (d) \text{ for some } d \in R.$$

Now, $d = ax + by$ for some $x, y \in R$, and, since $(a), (b) \subseteq (a, b) = (d)$, $d \mid a$ and $d \mid b$. And, if $d' \mid a$ and $d' \mid b$, then $d' \mid d (= ax + by)$. The converse is trivial.

Thm. Every finitely generated Ideal of a Bézout domain is principal.

Proof. Let R be a Bézout domain, and $I \subseteq R$ be a finitely generated Ideal. Denote

$$I = (a_1, \dots, a_n) \text{ for some } a_1, \dots, a_n \in R.$$

Using induction for n . If $n = 1$ or 2 , trivial. Suppose for $n-1$, the statement holds.

Then, $(a_1, \dots, a_{n-1}) = (p)$ for some $p \in R$. Enough to show $(a_1, \dots, a_n) = (p, a_n)$.

Let $a_1x_1 + \dots + a_nx_n \in (a_1, \dots, a_n)$ be given. Since for each a_i , $p \mid a_i$, thus

$$a_1x_1 + \dots + a_{n-1}x_{n-1} + a_nx_n = px + a_nx_n \in (p, a_n).$$

Converse is trivial. $\square$

Thm. If R is Bézout domain and also U.F.D., then it is P.I.D.

Proof. Let $I \subseteq R$ be an Ideal of R , and $a \in I$ be a nonzero, minimal number of irr. factors.

If $I \neq (a)$, then there is $b \in I \setminus (a)$. Since R is Bézout domain, for some $d \in I$,

$$(d) = (a, b) \subseteq I.$$

That is, $d \mid a$ and $d \mid b$. But, since the minimality of a , d must associates to a .

But, this implies $d \mid b \Rightarrow b \in (d) = (a)$, Contradiction. $\square$

Prop. In Bézout domain, every irreducible element is prime,
and, every prime ideal is Maximal.

Def. An Integral Domain R is called a Noether Domain if:

If a sequence of Ideals $\{I_n\}$ in R satisfies that:

there are no infinite increasing chains of Ideals.

Thm. Let R be an Integral domain.

R is Noether domain if and only if Every Ideal of R is finitely generated.

Proof. Suppose that R is Noether domain. Let $I \subseteq R$ be an infinite Ideal.

(If I is finite, then there is nothing to prove).

Let $a_1 \in I$ be given. If $(a_1) = I$, there is nothing to prove.

Let $a_2 \in I \setminus (a_1)$. If $(a_1, a_2) = I$, there is nothing to prove.

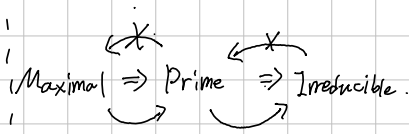
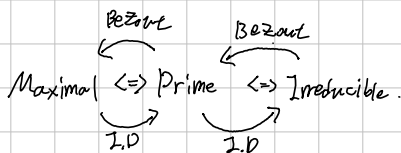
Inductively, $(a_1) \subsetneq (a_1, a_2) \subsetneq \dots \subseteq I$.

But, since R satisfies Ascending Chain Condition, for some $N \in \mathbb{N}$,

$$(a_1, \dots, a_N) = I.$$

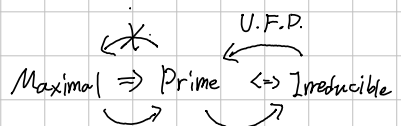
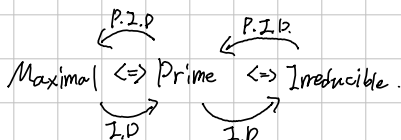
The converse is trivial.

Corollary. If R is Bézout domain and Noether domain, then R is Principal Ideal Domain.



Summary

Field $\Rightarrow$ Euclidean Domain $\Rightarrow$ Principal Ideal Domain $\Rightarrow$ Unique Factorization Domain $\subseteq$ Integral Domain.



Bézout domain

Noether domain $\Rightarrow$ Atomic domain.

If ir = prime.

Example. The polynomial ring $\mathbb{Z}[x]$ is Not P.I.D., but U.F.D.

Specifically, $(2, x) \subseteq \mathbb{Z}[x]$ is Not principal, but Maximal.

And, $(2) \subseteq \mathbb{Z}[x]$ is Irreducible and prime, but Not Maximal.

Example. The quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ is Not U.F.D., but Noether Domain.

Not U.F.D: $6 = 2 \cdot 3 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5})$ and

$2, 3, 1 \pm \sqrt{-5}$ are irreducible, because: Since unit of $\mathbb{Z}[\sqrt{-5}]$ is 1,

Example. Let $R = \mathbb{Z}[x]$. Then, $(2, x) \subset R$ has the following properties:

$(2, x) = \{2p(x) + xq(x) \mid p, q \in \mathbb{Z}[x]\} \subsetneq R$, since $1 \notin (2, x)$. Thus proper.

If $(2, x) = (a(x))$ for some $a(x) \in \mathbb{Z}[x]$, then, $2 \in (a(x))$, this means

$$2 = a(x) \cdot p(x) \text{ for some } p(x) \in \mathbb{Z}[x].$$

Now, $a(x), p(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $(a(x)) = (1) = R$, Contradiction. Thus $a(x) = \pm 2$. But, $x \notin (\pm 2)$, Contradiction. Now, $(2, x)$ is not principal.

But $(2, x) \subsetneq \mathbb{Z}[x]$ is Maximal, because a map

$$\varphi: \mathbb{Z}[x] \rightarrow \mathbb{Z}_2: p(x) \mapsto p(0) \bmod 2 \quad \mathbb{Z}[x] \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}_2.$$

is an onto Ring-Homomorphism having $\ker \varphi = (2, x)$ because

$$p(x) \in \ker \varphi \Leftrightarrow p(0) \bmod 2 = 0 \Leftrightarrow p(x) \in (2, x)$$

Thus $\mathbb{Z}[x]/\ker \varphi \cong \mathbb{Z}_2$ is a field, $\ker \varphi = (2, x)$ a Maximal, further, prime.

And, clearly, $2 \in R$ is irreducible but $(2) \subsetneq (2, x)$, it is prime, but not Maximal.

[Quadratic Integer Ring with Norm.

Def. Let D be a square free integer. Define

$$\mathbb{Z}[\sqrt{D}] \stackrel{\text{def}}{=} \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$$

is called a Quadratic Integer Ring. And, define a Norm on $\mathbb{Z}[\sqrt{D}]$ as

$$N: \mathbb{Z}[\sqrt{D}] \rightarrow \mathbb{N} \cup \{0\}; a + b\sqrt{D} \mapsto a^2 - Db^2$$

Since D is a squarefree,